20. Function Machines

Year 4 · Mark Scheme — Supplementary Worksheet

Foundation — Answers

1. 56

2. 21

3. 11

4. C — 7

5. 20

6. C — 19 (7 × 3 = 21, then 21 − 2 = 19)

7. 7

8. C — 7

9. 16

10. 24

11. A, B and D (3→12, 4→16 and 6→24 are all correct; C should be 5→20 not 19, and E should be 7→28 not 29)

12. 16

13. 8

14. D — 31 (8 × 3 = 24, then 24 + 7 = 31)

15. 0

Standard — Answers

16. 25

17. 24

18. 2

19. C — 4 (4 + 5 = 9, then 9 × 4 = 36)

20. 9

21. a = 3, b = 4 (from 3 to 6 is +3, and the output rises from 13 to 22, which is +9, so a = 9÷3 = 3; then 3×3 + b = 13, so b = 4)

22. 2

23. C — 9 (9 × 2 = 18, then 18 + 9 = 27)

24. 5

25. A, C and E (7→4, 11→12 and 15→20 are all correct; B should be 9→8 not 11, and D should be 13→16 not 17)

26. 6/7

27. 6

28. C — 8.5

29. 5

30. 2

31. 48

32. 28

33. 7

34. 11

35. 30

Hard — Answers

36. £26.50 (4 + 2.50 × 9)

37. 3 (4 × 5 = 20, so b = 23 − 20 = 3)

38. a = 3, b = 5 (from 2 to 5 is +3, and the output rises from 11 to 20, which is +9, so a = 9÷3 = 3; then 3×2 + b = 11, so b = 5)

39. 12 (7 × 3 = 21, then 21 − 9 = 12)

40. A, C and E (10→12, 14→14 and 18→16 are all correct; B should be 12→13 not 16, and D should be 16→15 not 14)

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